

Ex.No. 2: Implementing TDMA Slot Allocation in GSM. Date

Objective1

```
clc; clear; close all;
```

```
FrameTime = 4.615; Slots = 8;
```

```
SlotDuration = FrameTime/Slots;
```

```
t = 0:SlotDuration:FrameTime;
```

```
figure
```

```
subplot(2,1,1)
```

```
stem(t,ones(size(t)),'filled')
```

```
title('TDMA Time Slot Structure')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Level')
```

```
grid on
```

```
subplot(2,1,2)
```

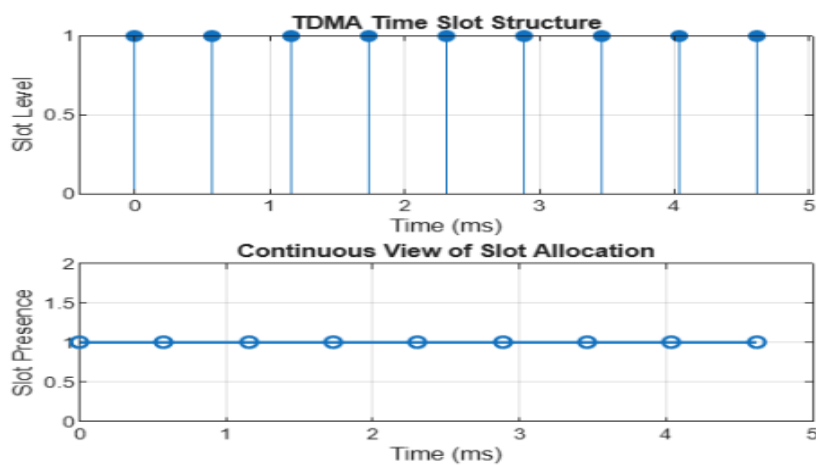
```
plot(t,ones(size(t)),'-o','LineWidth',1.5)
```

```
title('Continuous View of Slot Allocation')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Presence')
```

```
grid on
```



Objective2

```
clc; clear; close all;
```

```
TotalSlots = 8;
```

```
AllocatedSlots = 6;
```

```
FreeSlots = TotalSlots - AllocatedSlots;
```

```
Utilization = (AllocatedSlots/TotalSlots)*100;
```

```
disp('Frame Utilization (%)')
```

```
disp(Utilization)
```

```
figure
```

```
% Graph 1: Simple Line Plot (SAFE - no bar function)
```

```
subplot(2,1,1)
```

```
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
```

```
title('GSM Frame Utilization (Allocated vs Free)')
```

```
xlabel('Slot Type Index (1=Allocated, 2=Free)')
```

```
ylabel('Number of Slots')
```

```
grid on
```

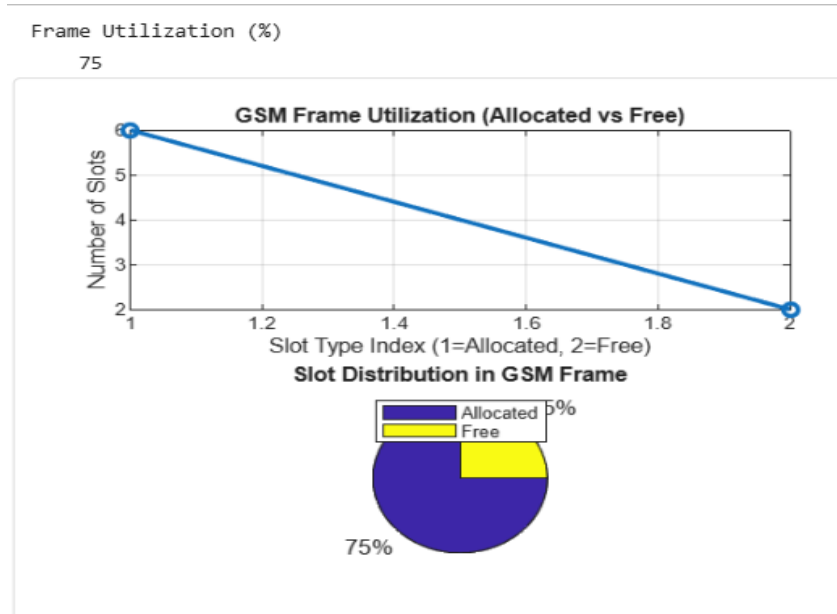
```
% Graph 2: Pie Chart (SAFE)
```

```
subplot(2,1,2)
```

```
pie([AllocatedSlots FreeSlots])
```

```
title('Slot Distribution in GSM Frame')
```

```
legend('Allocated','Free')
```



Objective 3

clc;

clear;

close all;

% TDMA Parameters

Users = 1:8; % Total Users

TotalSlots = 5; % Available Time Slots

% Slot Allocation

Alloc = zeros(1,8); % Initialize Allocation

Alloc(1:TotalSlots) = 1; % First 5 Users Allocated

Blocked = 1 - Alloc; % Remaining Users Blocked

% Display Allocation Table

disp('User Wise Slot Allocation');

disp(table(Users', Alloc', ...

'VariableNames', {'User', 'Status_Allocated_1_Blocked_0'}));

```
% Plot Graphs
```

```
figure;
```

```
% Graph 1: Slot Allocation
```

```
subplot(2,1,1);
```

```
stem(Users, Alloc, 'filled', 'LineWidth', 2);
```

```
title('TDMA Slot Allocation per User');
```

```
xlabel('User Index');
```

```
ylabel('Slot Status');
```

```
ylim([-0.2 1.2]);
```

```
grid on;
```

```
% Graph 2: Allocation vs Blocked
```

```
subplot(2,1,2);
```

```
plot(Users, Alloc, '-o', 'LineWidth', 2);
```

```
hold on;
```

```
plot(Users, Blocked, '-s', 'LineWidth', 2);
```

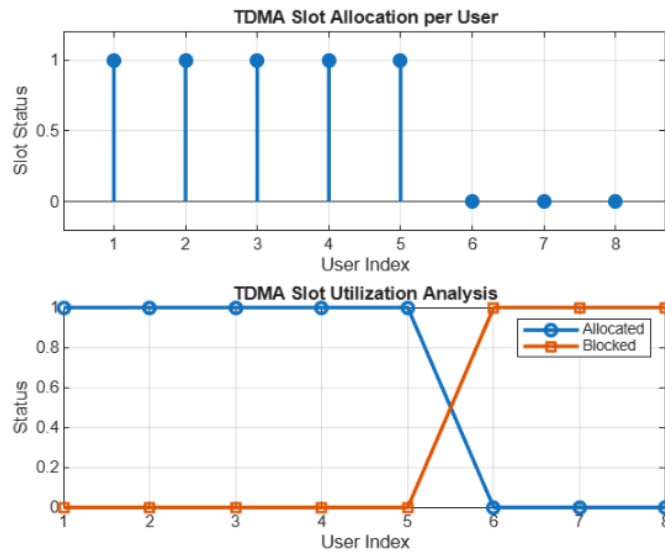
```
hold off;
```

```
title('TDMA Slot Utilization Analysis');
```

```
xlabel('User Index');
```

```
ylabel('Status');
```

```
legend('Allocated', 'Blocked');
```



grid on;